

SEPM ASSIGNMENT 2

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Q1 Define a project? what is project management? Explain various characteristics of project management

A Project

A project is a temporary activity undertaken to create a unique product, service or result within a defined time, cost and scope.

Ex:- Developing a banking software system

Project Management

Project Management is the application of knowledge, skills, tools and techniques to project activities to meet project requirements.

Characteristics of Project Management

- a) Temporary Nature - Has a definite start and end
- b) Unique Output - Every project delivers something new
- c) Defined Objectives - clear goals must be achieved
- d) Resource Constraints - limited time, cost, manpower
- e) Interdependent Tasks - Tasks depend on each other
- f) Risk and uncertainty - Project involves risk
- g) Progressive Elaboration - Plans are refined over time

f) Multiple people are involved
(Clients, team, managers)

Q2 Write a short note on PMBOK?

A PMBOK (Project Management Body of Knowledge) is a standard guide published by PMI

- Provides best practices in project management
- Define processes, tools and techniques
- widely used locally

a Process Groups

These represent the lifecycle ~~staf~~ stages of a project.

1. Initiating - Defines the project at a high level. Identify stakeholders and create project charter.
2. Planning - Defines what, how, when and ~~or~~ who. Prepare detailed plan of Scope, cost and schedule. It includes scope planning, risk planning, resource planning
3. Executing - Actual work is performed. Team members complete assigned tasks. It includes task execution, team management, communication

8. Risk Management - Reduces uncertainties and manages risks
9. Procurement Management - Purchasing goods/Services, managing vendors/contracts.
10. Stakeholder Management - Identifying stakeholders, managing their expectations and engagement

Q3 Explain ITPM

A ITPM (Information Technology Project Management) is the process of planning, executing, monitoring and controlling IT-related projects such as software development, ^{net}working, database systems and cybersecurity solutions.

It combines Project management principles and Technical IT knowledge.

Objectives of ITPM

- Deliver IT Solutions on time.
- Maintain quality standards.
- Control cost and resources.
- Ensure customer segmentation.
- Adapt to changing requirements.

Phases of ITPM

Initiation → Planning → Execution
→ Monitoring & Control → Closure

Monitoring and Controlling - Tracks project performance - compare actual vs planned and includes performance measurement, risk monitoring, change control. Output includes correction & updates

5. Closing - Final completion of project. Deliver product to client. It includes final documentation, client approval and team release.

6 Knowledge Areas

These are the core areas of project management knowledge

1. Integration Management - Combines all project activities
2. Scope Management - Prevents scope creep and defines what is included in the project
3. Time (Schedule) Management - Ensures timely completion
4. Cost Management - Budget estimation and control
5. Quality Management - Focus on quality assurance & control
6. Human Resource (HR) Management - Managing team members and assigning responsibilities
7. Communication Management - Ensures proper flow of information

Q3

Tools and Techniques used in ITPM

- Gantt charts → for scheduling
- PERT/CPM → for project planning
- Agile/scrum → for software development
- Project management software

Example → Developing a Banking Management System

Planning → define features

Execution → coding modules

Monitoring → testing and debugging

Closure → deploy system in bank

Q4 What is the role of RFP? Describe its content and purpose?

A RFP (Request for Proposal) is a document issued by an organization to invite vendors to submit proposals.

Role of RFP

- Helps in selecting the best vendor
- Defines project requirements clearly
- Ensure fair competition

Contents of RFP

- Project overview
- Scope of work
- Technical Requirements
- Budget Details

Timeline

Evaluation Criteria

Terms & Conditions

Purpose

- To communicate requirements
- To compare vendor proposals
- To select suitable contractor

Q5 what is feasibility study? Explain its types, contents and purpose?

A A feasibility study determines whether a project is practical and achievable

Types of feasibility

- Technical Feasibility
Is the technology available
- Economic Feasibility
Is it cost effective
- Operational Feasibility
Will it work in the organization
- Legal Feasibility
Any legal issues
- Scheduled Feasibility
Can it be implemented on time

Contents

- Problem Definition
- ~~Proposed~~ Proposed Solution
- Cost-Benefit Analysis
- Risk Analysis
- Resource Requirements

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Purpose:

- Avoid project failures
- Supports decision making
- Ensure efficient resource use

Q6 Write a short note on W.B.S.

A WBS stands for work Breakdown structure

WBS is a hierarchical decomposition of a project into smaller tasks

Breaks projects into manageable units
Improves planning and control
Helps in cost and time estimation

Advantages

- Better task assignment
- Clear project structure
- Easier monitoring.

Q7 Consider an XYZ Company under a take a project to computerized working of ABC City Bank, Then develop WBS.

A Work Breakdown structure is a hierarchical decomposition of a project into smaller tasks

Breaks project into manageable units and improves planning and control

Project : Computerization of ABC City Bank

Level 1 : Project

- Banking System Development

Level 2 : Major Phases

① Requirement Analysis

- Data collection
- Stakeholder meetings
- Requirement documentation

② System Design

- Architectural Design • UI Design
- Database Design

③ Development

- Frontend & Backend development
- API Integration

④ Testing

- Unit testing • Integration testing
- System testing

⑤ Implementation

- Installation
- Data migration
- User training

⑥ Maintenance

- Bug fixing • updates
- Performance monitoring

Level 3 : (Example Breakdown)
Development → Backend

- Account management module,
- Transaction processing • security